MECHANICAL - WAREHOUSING & MHE SYSTEMS

DRAFT

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SECTION-I
GENERAL INSTRUCTIONS

GENERAL INSTRUCTIONS

1.1 Purpose of this document is to provide Technical Scope of Work information for the EPC Contract bidder to prepare technical and commercial proposal for the Scope of Design, Engineering, Procurement, Supply, Erection / Installation, Painting, Testing, Pre-commissioning, running trials and Commissioning of Automated Storage and Retrieval system based Warehouses with Warehouse Management System in the back-end, Trace & Track system and Material Handling systems – for Raw Material warehouse, Packaging Material warehouse, Finished product warehouse and Low Flash storage warehouses including integration with ERP system, TAS system, ACS system, CCTV system, FF systems, Safety system and other related systems - as part of Mechanical Package #2F for Indian Oil Corporation Limited's Grass-root Lube Complex Project at Ammulavauyol and Vaikkadu, Chennai, India. The scope also includes procurement, supply of suitably designed pallets in steel having suitable tagging arrangement – for handling in warehouses.

1.2 This Lube complex shall be designed to be fully automated plant, right from receipt of raw materials or packaging materials, storage of raw materials in automated warehouses or bulk storage tanks, issue/transfer of raw materials to Process, automated processing, Transfer of Finished product to bulk FP storage, Transfer of Bulk products to filling lines or Tank-truck filling gantry for bulk TT supplies, automated filling & packaging operations, transfer of packed products from filling halls to storage in FP warehouse and finally loading & dispatch of packed finished products.

1.3 Contractor scope includes Design, Engineering, Procurement & Supply, and Site Erection & Commissioning Services – including integration / interfacing with other support systems, stated above.

1.4 The details provided in this document are basic minimum requirements. The EPC Contractor, with domain expertise in offering Warehousing solutions, shall build upon this and suggest the most appropriate integrated system for the application keeping in view the philosophy for a completely automated warehousing solution.

EPC contractor shall confirm full compliance or list out any deviations which they feel are necessary with this Scope of Works, attached documents and applicable local regulations and

codes. Reference to standards and codes in this Scope of work shall in no way relieve the EPC Contractor of the responsibility to comply with all requirements.

1.5 Contractor shall be responsible for design, procurement, installation and commissioning of completely automated Warehouses including integration with existing SAP with storage solution and state of the art automated material handling equipment.

1.6 The Warehouse shall be designed with automated storage solution along-with compatible track and trace system for identification, date stamping & quantification of materials received / dispatched – to ensure seamless integration for material transfer operations.

1.7 WMS software shall be provided with back-up provision of Application Program, data transactions and help menu in program manual.

1.8 Overall system shall be designed considering the operations in Warehouse (raw materials, packing materials and finished products) design including Warehouse architecture, layout, logistics, Storage solution including racking / other storage equipment, material handling equipment duly integrated with WMS software.

1.9 Truck loading for barrels and smalls with docking station arrangements shall be provided. Hydraulically operated Dock Levellers shall be supplied to connect the floor and the container truck to ensure forklift movement inside the container for ease of loading. Scope also includes design of suitable material conveying arrangements viz. Telescopic conveyors for loading / unloading of goods into the truck /out from the trucks.

1.10 Providing a Ware House inside the Lube Complex will help in effective utilization of storage space requirement for material storage, better control, economy in automated centralized operations instead of separate Warehouses in respective manufacturing sheds. Also it will help in better control on movements of packed trucks inside the plant.

1.11 Contractor shall establish the feasible and economical warehouse system for Raw Material, Packaging Material and Finished Product warehouses.

1.12 Warehouse for Low Flash Materials being smaller and also as it is necessary to have adequate access during any fire situation, it is proposed to provide only a regular Random access

warehouse with Fork lift truck / Stackers as MHEs.

1.13 Packaging Material warehouse will be used for storing empty packaging materials except Drums which will be manufactured and stored in Drum Silos.

1.14 EPC Contractor shall be responsible for selection of systems specified in Datasheets and Specification. EPC Contractor shall evaluate quotations from Vendors and submit Technical Bid Analysis report for review by Owner / Consultant and also submit the unpriced PO.

1.15 All materials required for the Scope of Work as well as for temporary and enabling work shall be arranged and supplied by EPC Contractor. All incidental expenses, loading, unloading, transportation, handling etc. shall be the responsibility of the EPC Contractor and cost towards such expenses should be included in the finished item rates.

1.16 EPC Contractor shall provide their quality assurance system and quality control and inspection of materials and workmanship to ensure that the requirements of scope of work, specifications and codes are met.

1.17 All design and construction of Civil works including building, wall / cladding, supports, etc., are excluded from the scope of EPC Contractor.

1.18 EPC Contractor shall co-ordinate with the Owner / Consultant and provide information and inputs required for the applicable foundation and structural design and installation. EPC Contractor shall also coordinate with the Owner / Consultant to facilitate timely installation of foundation and supports structure by Other Contractors.

1.19 Shop Inspections of items purchased by EPC Contractor will be carried out by Owner Appointed Third Party Inspector (TPI) wherever deemed necessary.

1.20 TPI access to shops and advanced notifications for shop inspections shall be managed by EPC Contractor. EPC Contractor shall submit a detailed Inspection & Testing Plan for all the material purchased by it for approval along with bid and also a final ITP prior to commencement of Procurement & Fabrication for Owner review & approval.

1.21 Any accessories required to make the equipment complete, safe and consistent with good engineering practice shall be provided by the EPC Contractor at no extra cost and is deemed

to have been specified.

1.22 EPC Contractor shall ensure that the equipment listed in this Tender Scope of Works are designed and constructed in accordance with government statutory requirements.

1.23 In the event of any conflict in requirement, the following order of precedence shall apply:

a. Local Regulation and Statutory Requirement.

b. Equipment Specifications and Datasheets

c. Project Specifications

d. Local and International Codes and Standards

Conflicts or ambiguities among the referenced codes, standards, etc. and this Tender Scope of Works shall be immediately referred to the Consultant / Owner by Contractor in writing for resolution. As a general rule, the most stringent requirement shall govern. Local code & statutory requirement shall be complied with even if it is not specifically stated.

1.24 EPC Contractor is free to propose alternative design based on their experience for purpose of reducing project cost and improvement of operability. All proposed deviations to the Specifications shall be approved by the Owner / Consultant in advance. Contractor is responsible to identify and raise deviations to Specifications referenced.

1.25 EPC Contractor shall make no change to, or deviate from, the requirements of the specification, except by prior written permission of Owner / Consultant.

1.26 Information provided in this document is intended to provide a starting point for the EPC Contractor to develop the final optimum design. EPC contractor shall perform FEED verification on these documents, update and / or develop design documentation as necessary.

1.27 EPC Contractor shall update equipment design basis, specifications, drawings, datasheets, equipment list, etc. and also prepare required datasheets / drawings, where not available.

1.28 Approval of Contractor drawings, calculation and other documents does not relieve the Contractor’s responsibility of complying with this requisition and its attached project

specification, datasheet, drawing etc.

1.29 Design alternatives including material changes, equipment sparing and built-in maintenance enhancements, design enhancements to save the capital and operating cost shall be developed and proposed for the Owner / Consultant review and approval.

1.30 EPC Contractor shall interface and co-ordinate with Civil EPC Contractor per agreed interface procedure and provide information and inputs required for foundation design and construction. EPC Contractor shall also coordinate with other respective EPC contractors to facilitate timely inputs, design, construction and installation.

1.31 EPC Contractor shall record and maintain records of the quality control inspection activities performed and have system to segregate non-conforming products or workmanship. A list of any non-conformances shall be issued on a regular basis.

SECTION-II
DETAILED SUPPLY AND SERVICES SCOPE

DETAILED SUPPLY AND SERVICES SCOPE

1.1 This section covers the minimum requirements of supply, procurement, erection/ installation, testing, pre-commissioning & commissioning including complete software integration – for complete Warehouse Management Solution – comprising of appropriate Storage racking system, Automated storage & retrieval system, warehouse management system, trace & track system including associated Material Handling equipment (viz. Fork lift trucks, Stackers, Reach Trucks, Conveyors, etc), Sensors & Coders etc.

The specification covers the scope of work of the contractor to be carried out, general guidelines, codes, standards and general engineering practice to be followed.

1.2 The scope of work shall be read in conjunction with Schedule of Rate (SOR), technical specifications and datasheets enclosed with this tender document.

1.3 Contractor shall submit detailed design datasheet and drawings based on the basic design data provided by Owner/Consultant with OEM's confirmation on product life cycle and their support for backup engineering & spares.

1.4 Any other equipment / service, which is not explicitly mentioned in this document or in the Schedule of Quantity / price bid but deemed necessary for the successful operation of the package unit complete in all respects, shall be in EPC Contractor's scope.

1.5 The functional & operational requirements as specified in this tender might undergo revisions as per the process and / or fluid requirement during detail engineering, functional design specification, and factory acceptance test or during execution. Contractor & Package Vendor shall ensure that all these changes are incorporated & demonstrated during Site Acceptance Test.

1.6 All technical personnel assigned to the site shall be fully conversant with the supplied system and shall have necessary capability to bring the system on line quickly and efficiently and providing training to Owner team as regards to Operations & Maintenance of Packages.

1.7 Performance testing of the completely assembled / erected package unit as a whole – to establish the guarantee parameters specified in order.

1.8 The scope of work of the Contractor shall include but not limited to the following:

A Detailed Design and Engineering, Procurement, Supply, Installation, Erection and Commissioning including preparation and submission of documents (General Arrangement, detail drawings, as-built drawings, calculations, etc.,) providing necessary accessories, spares, special tools & tackles, testing, calibration, painting etc. as per drawings , technical specifications and applicable codes and standards. The scope shall include complete ASRS

racking, Pallets, associated aisle equipment including power bus bar / flexible cable, top rail, bottom rail, conveyor systems for pallets loading / unloading, suitable stacker cranes in required quantities for each ASRS, telescopic fork, suitable transfer vehicles or cars in required quantities, all necessary controls, PLC, IT hardware & software for Automation, supply, fabrication & installation as per drawings and specification shall also be in the scope of bidder. Painting (external surface, support structure etc.) of ASRS shall be carried out as per the approved painting procedure.

a. One (1) number Raw Material AS/RS for approximate dimensions 31.3 m wide x 51.1 m length x 12 m height conforming to FEM standards 9.311, 9.831, 9.851 & 9.881.

b. One (1) number Packaging Material AS/RS for approximate dimensions 23.2 m wide x 75.0 m length x 12 m height conforming to FEM standards 9.311, 9.831, 9.851 & 9.881.

c. One (1) number Finished Product AS/RS for approximate dimensions 38.2 m wide x 75.0 m length x 12 m height conforming to FEM standards 9.311, 9.831, 9.851 & 9.881.

S.No	Package Description	Qty
1	RAW MATERIAL WAREHOUSE – AS/RS	1
	Pallet Quantity: 2500 numbers	
2	PACKAGING MATERIAL WAREHOUSE – AS/RS	1
	Pallet Quantity: 1500 numbers	
3	FINISHED PRODUCT WAREHOUSE – AS/RS	1
	Pallet Quantity: 4500 numbers	

Note:

1. *EPC Contractor shall engage with reputed ASRS suppliers to optimize following warehouse storage type / details such as, (i) single deep vs double deep system, (ii) dedicated automated linear stacker crane for each aisles vs combination of automated linear stacker crane with transfer car options. Optimisation shall be based on operation cycles specified in the attached reference documents for ASRS.*

   Storage Racks shall be designed for Pallet sizes of 1200 mm x 1200mm x 200mm height steel pallets of suitable design arranged in 6 tiers; Each loaded pallet outer dimensions are considered as 1200mm x 1200mm x 1300mm and loaded pallet weight is considered as 1150 kgs UDL (including self weight of the Pallet). Vendors to validate these design considerations.

2. *Vendor design to also consider the requirement for provision of In-rack sprinkler for*

the Storage racks – in accordance with guidelines of NBC / NFPA and accordingly suitable free-board space shall be considered during finalization of rack-to-rack vertical spacing.

3. *Suitable Stretch wrapping or Strapping provision shall be included in the scope of Vendor for Loaded pallets carrying Cartons or packaging materials etc. suitably to avoid any fall of items during handling, before sending to the ASRS.*

4. *Office and Server room and furniture for these rooms (including UPS of adequate rating) will also be in the scope of the Vendor.*

5. *Only Main electrical connections to the Main Electrical Power distribution panel and LAN network shall be provided by client at 1 point for each warehouse.*

6. *Warehouse Management System shall include Inventory Control, Location Management & Handling Control Software modules and Integration with ERP, TAS & Filling PLC.*

7. *Contractor to include tertiary barcode / label printing, mobile data entry (MDE) & stretch wrapping and Finished Product pallet identification.*

8. *Contractor to include RFID system, MDE & stretch wrapping for Additive Raw Material & Packaging Material pallet identification.*

9. *UPS back-up for WMS Server and PLC operation shall be provided by EPC Contractor.*

B Detailed Design and Engineering, Procurement, Supply, Installation and Erection including preparation and submission of documents (General Arrangement, detail drawings, as-built drawings, calculations, etc.,) providing necessary accessories, special tools, testing, calibration, painting etc. as per drawings , technical specifications and applicable codes and standards – for Random Access Rack Warehouse for Low Flash raw materials, associated aisle equipment for pallets loading / unloading, track-n-trace system, integration with WMS as per drawings and specification shall also be in the scope of bidder. Painting (external surface, support structure etc.) shall be carried out as per the approved painting procedure.

a. One (1) number Low Flash Raw Material Random Access Storage Rack Warehouse for approximate dimensions 14 m wide x 35 m length x 4 m height confirming to FEM standards 10.2.02, 10.3.01, BS EN 15512 & SEMA Practices.

.No	Package Description	Qty
1	LOW-FLASH RAW MATERIAL WAREHOUSE RACK	1
	Pallet Quantity: 150 numbers; Two (2) level, 24 SKUs	

C Detailed Design and Engineering, Procurement, Supply, Installation and Erection including

preparation and submission of documents (General Arrangement, detail drawings, as-built drawings, calculations, etc.,) providing necessary accessories, special tools, testing, calibration, painting etc. as per drawings , technical specifications and applicable codes and standards. The scope shall include Dock Levellers and hinged lip assembly with associated power pack equipment, suitable power accessories with push-button controls in required quantities, Operator Control Unit with hardware in IP65 enclosure, Heavy duty non-skid Platform, Lift Cylinder with Safety Stop Valve, Super Bumpers. Supply, fabrication & installation as per drawings and specification shall also be in the scope of bidder. Painting (external surface, support structure etc.) of Dock Levellers shall be carried out as per the approved painting procedure.

a. Four (4) numbers of Dock Levellers for Raw Material Loading & unloading with approximate dimensions 3.0 m wide x 3.0 m length x 0.55 m height confirming to EN standard 1398. Lip Extension shall be 0.41 m.

b. Four (4) numbers of Dock Levellers for Packaging Material Loading & unloading with approximate dimensions 3.0 m wide x 3.0 m length x 0.55 m height confirming to EN standard 1398. Lip Extension shall be 0.41 m.

c. Nine (9) numbers of Dock Levellers for Finished Product Loading & unloading with approximate dimensions 3.0 m wide x 3.0 m length x 0.55 m height confirming to EN standard 1398. Lip Extension shall be 0.41 m.

d. Eight (8) numbers of Dock Levellers for Empty Drum Loading & unloading with approximate dimensions 3.0 m wide x 3.0 m length x 0.55 m height confirming to EN standard 1398. Lip Extension shall be 0.41 m.

S.No	Package Description	Qty
1	DOCK LEVELLERS FOR RAW MATERIAL UNLOADING AREA	4
2	DOCK LEVELLERS FOR PACKAGING MATERIAL UNLOADING AREA	4
3	DOCK LEVELLERS FOR FINISHED PRODUCTS UNLOADING AREA	9
4	DOCK LEVELLERS FOR DRUM PLANT	8

D Detailed Design and Engineering, Procurement, Supply, Installation and Erection, Testing, Commissioning including preparation and submission of documents (General Arrangement, detail drawings, as-built drawings, calculations, etc.,), providing necessary accessories, special tools, testing, calibration, painting etc. as per drawings, technical specifications and applicable codes and standards. The scope shall include Goods Lifts MRL type, Car Cabin complete with safety device, brakes and buffer springs, Hoist motor with variable frequency control drive, Guide Rails with guide shoes, Counter Weights, Hoisting & Governor Ropes, Door operators complete with variable drive motors, Emergency Stops, Maintenance

Switches, Interlocks & Overload indicators. Supply, fabrication, installation & testing as per drawings and specification shall also be in the scope of bidder. Painting shall be carried out as per the approved painting procedure.

a. Three (3) numbers of Goods Lifts MRL type for Material Handling 2MT Capacity with approximate dimensions 2.2 m wide x 2.8 m length x 2.2 m height confirming to IS standard 14665.

b. Obtain Statutory Approval from the Inspectorate of Lifts as may be required for commissioning of the lifts and handover for operation after satisfactory tests. Contractor shall follow all Statutory Requirements as well as best trade practices in the manufacture & installation of lifts.

S.No	Package Description	Qty
1	GOODS LIFT (CAPACITY: 2 MT)	3

E Detailed Design and Engineering, Procurement, Testing and Supply including preparation and submission of documents (General Arrangement, detail drawings, calculations, etc.,), providing necessary accessories, special tools, testing, calibration, painting etc. as per drawings, technical specifications and applicable codes and standards. The scope shall include Electric Battery Operated Fork Lift Trucks, complete with Power Steering, 3 stage lift mast, Load back rests, Suitable Batteries, Battery Chargers, Discharge Indicators, Lights, Standard Tools, etc. Supply, fabrication, installation & testing as per drawings and specification shall also be in the scope of bidder. Painting shall be carried out as per the approved painting procedure. Following are under Contractor's scope of supply & services:

a. Six (6) numbers of Battery Operated Fork Lift Trucks for Pallets & Material Handling with approximate 2 MT capacity.

S.No	Package Description	Qty
1	FORK LIFT TRUCKS (BATTERY POWERED)	6

F Detailed Design and Engineering, Procurement, Testing and Supply including preparation and submission of documents (General Arrangement, detail drawings, calculations, etc.,), providing necessary accessories, special tools, testing, calibration, painting etc. as per drawings, technical specifications and applicable codes and standards. The scope shall include Diesel Powered Fork Lift Trucks, complete with Power Steering, 3 stage lift mast, Load back rests, Suitable Batteries, Lights, Standard Tools, etc. Supply, fabrication, installation & testing as per drawings and specification shall also be in the scope of bidder. Painting shall be carried out as per the approved painting procedure. Following are under Contractor's scope of supply

& services:

a. Four (4) numbers of Diesel Powered Fork Lift Trucks for Pallets & Material Handling with approximate 2 MT capacity.

S.No	Package Description	Qty
1	FORK LIFT TRUCKS (DIESEL POWERED)	4

G Detailed Design and Engineering, Procurement, Testing and Supply including preparation and submission of documents (General Arrangement, detail drawings, calculations, etc.,), providing necessary accessories, special tools, testing, calibration, painting etc. as per drawings, technical specifications and applicable codes and standards. The scope shall include Electric Pallet Trucks, complete with Power Steering, 3 stage lift mast, Load back rests, Suitable Batteries, Lights, Standard Tools, etc. Supply, fabrication, installation & testing as per drawings and specification shall also be in the scope of bidder. Painting shall be carried out as per the approved painting procedure. Following are under Contractor's scope of supply & services:

a. Six (6) numbers of Electric Pallet Trucks with approximate 2 MT capacity.

S.No	Package Description	Qty
1	PALLET TRUCKS (BATTERY POWERED)	6

H TRACK-N-TRACE SYSTEM

Detailed Design and Engineering, Procurement, Testing and Supply including preparation and submission of documents (General Arrangement, detail drawings, calculations, etc.,), providing necessary accessories, special tools, testing, calibration etc. installation, pre-commissioning, running trials, co-ordination with Filling line End-of-Line systems, integration with WMS and CFE, as per drawings, technical specifications and applicable codes and standards. The scope shall include required provision of required no & models of Scanners, Readers, Coders etc. including associated conveyors / Handling equipments for effective operation – at all receiving / transfer points (viz. End of filling lines, Additive & Packaging material receiving docks) and issuing points (Additive & Packaging material issuing points) and FP loading points – generally as per the Functional Design Requirements as specified in the attached document. Following are under Contractor's scope of supply & services:

Track-N-Trace systems for all 4 Warehouses :

S.No	**Package Description**	**Qty**

1	Track-N-Trace system for Raw material Warehouse	1 set
2	Track-N-Trace system for Packaging material Warehouse	1 set
3	Track-N-Trace system for Finished product Warehousing system	1 set
4	Track-N-Trace system for Low Flash storage Warehouse	1 set

I LOADING / UNLOADING / CONVEYORS SYSTEMS

Detailed Design and Engineering, Procurement, Testing and Supply including preparation and submission of documents (General Arrangement, detail drawings, calculations, etc.,), providing necessary accessories, special tools, testing, installation, pre-commissioning, running trials, commissioning - as per drawings, technical specifications and applicable codes and standards. The scope shall include required provision of TELESCOPIC CONVEYORS OR ROUGH TOP BELT CONVEYORS OR EQUIVALENT LOADING / UNLOADING SYSTEMS of suitable capacity and in required quantities - of reputed make, including associated conveyors / feeding equipment for effective operation – at all loading / unloading points – generally as per the Functional Design Requirements. Following are under Contractor's scope of supply & services:

Telescopic conveyor systems for all 4 Warehouses :

S.No	Package Description	Qty
1	MHEs for Loading of Packed products – for FP warehouse	As required
2	MHEs for Unloading of Additive Drums / Powder bags – for Additive Warehouse	As required
3	MHEs for Unloading of Packaging materials from Trucks – for PM Warehouse	As required
4	MHEs for Unloading of Low flash additives in Low Flash warehouse	As required

J EMPTY DRUM TRANSFER SYSTEM – FOR GREASE & BITUMINOUS LUBE DRUMS

Detailed Design and Engineering, Procurement, Testing and Supply including preparation and submission of documents (General Arrangement, detail drawings, calculations, etc.,), providing necessary accessories, special tools, testing, installation, pre-commissioning, running trials, commissioning - as per drawings, technical specifications and applicable codes and standards. The scope shall include provision of ELEVATED DRUM CONVEYOR FOR EMPTY GREASE / BITUMINOUS LUBE DRUMS of suitable capacity and design and reputed make, suitable for conveying empty Drums from Drum Plant to Grease /Bituminous Lube plant,

including associated feeding & discharge equipment for effective operation – at loading / unloading points. Following are under Contractor's scope of supply & services:

S.No	Package Description	Qty
1	Elevated empty Drum conveyor – for conveying of empty drums from Drum plant (0.3M Level) to Grease filling section (7.0M Level)	1no

K **STEEL PALLETS**

Detailed Design and Engineering, Procurement, Fabrication, Testing and Supply including preparation and submission of documents (General Arrangement, detail drawings, calculations, etc.,), - as per drawings, technical specifications and applicable codes and standards. The scope shall include provision of STEEL PALLETS OF SIZE 1200 X 1200MM X overall height 200mm; suitable for storage & handling of storage load upto 1150kgs (including self weight), of reputed make.

S.No	Package Description	Qty
1	Pallet type – 4 way, Non-reversible	8,650 no

L SAFETY SIGNAGES & BARRIERS

Design, procurement, supply, installation / mounting of signages & barriers that shall be provided for safety of warehouse personnel and longer life of the storage system. Quantity of signage will be based on the layout and as per statutory guidelines viz. OSHA. Following are the signages envisaged :

- EXIT indicators
- BAY / AREA signages (including Part Pallets area, Empty Pallet area etc.)
- SAFETY signages
- KEEP CLEAR signs
- LOADING DOCK signs
- PPE Signs
- WARNING – BATTERIES BEING CHARGED
- TRUCK SAFETY Labels
- INSTRUCTIONS for DRIVERS (Drivers restricted to Loading deck or SHUT OFF Engine while parked at Dock etc.)

1.9. **VENDOR DATA REQUIREMENT**

- **With proposal: -**

o Complete Technical details of the machines & accessories offered with leaflets, brochures, catalogs, photographs etc.

o Comments on our technical specifications, terms & conditions clearly indicating acceptance or deviations if any.

o The Bidder shall provide with his bid an overall "standard" inspection and quality assurance plan for all major equipment parts, including the following:

- Test and checks on major and sub-components.
- Certifying documents for raw materials, material tests and inspections, including certification level.
- Acceptance criteria.
- Inspection levels (witnessed/un-witnessed/documentation review, etc.)

o All data sheets, drawings and technical documentation for the package unit components – duly filled in with vendor data as required;

o Control schematic and control narrative, including details of WMS SYSTEM with Hardware and Software details, integration with IOCL's SAP system;

o List of recommended commissioning spares included in offer.

o List of Vendor's standard spares parts for 'two' years normal operation. This list shall be made separately for each item including auxiliaries and drivers in the form of table and show: -Part name, description and number. -Quantity installed in one unit -Quantity recommended per unit for two years normal operation

o An itemised list of special tools included in the offer.

o A specific statement that the scope of supply, the offered equipment, system and all its components are in strict accordance with data sheets, job specifications, this specification and all documents, except for specific deviations as listed in the proposal.

o Duly filled in experience Track record - giving details of Models, operating parameters, name of purchaser, consultant, date of commissioning and number of operating hours completed.

- **After order placement: -**

Drawings and data to be furnished by the Bidder /Vendor for Company's review / approval as outlined below shall be submitted as a part of Equipment Data Book.

- Schedule for the job execution

- Quality Assurance, Inspection & test Plan – for approval
- Inspection & Test procedures – for approval
- General arrangement drawing for the unit and all its accessories (final – good for construction) showing full details in plan and sections;
- Instrumentation and Operational control scheme – with detailed control narrative; Control logic with respect to sequence of operations;
- Cause & Effect drawings
- Detailed Installation drawing and static load details
- Cross-sectional drawing with BOM
- Technical Specifications & Datasheets for bought out items along with details of makes
- As built drawings (datasheet and dimensional drawings)
- Material Test certificates – for critical items
- Original Manufacturers' Test certificates – for all major bought out items
- Installation Manual
- Operation, Maintenance & Safety Manual
- Technical Data Manual / Mechanical catalogue
- Manufacturer's Performance guarantee / Warranty certificate
- Process Performance Guarantee certificate
- The Vendor shall provide completely filled-in data sheets, first for "As Purchased" and then for "As Built". This shall be done by the vendor, correcting and filling out the data sheets and submitting copies, to purchaser.

All documentation and drawings shall be filed as "FINAL DOCUMENTATION DOSSIER" in the form of hard copy and soft copy. 4 Hard copies in hard covered file & 2 soft copies in the form of CDs / DVDs.

1.10 Major Equipment under Scope of Supply of EPC Contractor:

S.No	Package Description	Qty	Remarks
A1	**RAW MATERIAL WAREHOUSE – AS/RS**	1	
	Pallet Quantity: 2500 numbers		
	Six (6) level, 230 SKUs		
A2	**PACKAGING MATERIAL WAREHOUSE – AS/RS**	1	
	Pallet Quantity: 1500 numbers		
	Six (6) level, 100 SKUs		
A3	**FINISHED PRODUCT WAREHOUSE – AS/RS**	1	
	Pallet Quantity: 4500 numbers		
	Six (6) level, 533 SKUs		
B	**LOW-FLASH RAW MATERIAL WAREHOUSE RACK**	1	
	Pallet Quantity: 150 numbers		
	Two (2) level, 24 SKUs		
C1	**DOCK LEVELLERS FOR RAW MATERIAL UNLOADING AREA**	**4**	
C2	**DOCK LEVELLERS FOR PACKAGING MATERIAL UNLOADING AREA**	**4**	
C3	**DOCK LEVELLERS FOR FINISHED PROUCTS UNLOADING AREA**	**9**	
C4	**DOCK LEVELLERS FOR DRUM PLANT**	**8**	
D	**GOODS LIFT**	**3**	
	Capacity: 2 MT		
E	**FORK LIFT TRUCKS (BATTERY POWERED)**	6	
F	**FORK LIFT TRUCKS (DIESEL POWERED)**	4	

G	**PALLET TRUCKS (BATTERY POWERED)**	6	
H1	**TRACK-N-TRACE SYSTEM FOR RM W/H**	1 SET	
H2	**TRACK-N-TRACE SYSTEM FOR PM W/H**	1 SET	
H3	**TRACK-N-TRACE SYSTEM FOR FP W/H**	1 SET	
I1	**MHE FOR LOADING /UNLOADING OF TRUCK (TELESCOPIC CONVEYOR OR ROUGH TOP BELT CONVEYOR) – FOR RM WAREHOUSE**	1 SET	
I2	**MHE FOR LOADING /UNLOADING OF TRUCK (TELESCOPIC CONVEYOR OR ROUGH TOP BELT CONVEYOR) – FOR PM WAREHOUSE**	1 SET	
I3	**MHE FOR LOADING /UNLOADING OF TRUCK (TELESCOPIC CONVEYOR OR ROUGH TOP BELT CONVEYOR) – FOR FP WAREHOUSE**	1 SET	
J	**EMPTY DRUM TRANSFER SYSTEM**	1 SET	
K	**STEEL PALLETS**	8,650 nos	
L	**SIGNAGES**	As reqd.	

SECTION-III
LOCAL REGULATORY CODES AND STANDARDS

LOCAL REGULATORY CODES AND STANDARDS

1.1 EPC Contractor shall comply with the requirement of following relevant Indian Statutory Codes and Standards including without limitation:

a. Factories Act

b. Indian Petroleum Rules

c. National Building Code (Latest)

d. Civil Aviation Rules

e. Indian Boiler Regulation Act

f. India Electricity Rules

g. Requirement of Chief Controller of Explosives

h. OISD (Oil Industry Safety Directorate) requirements

i. Environment Protect Act

j. SMPV Rules

k. Requirement of Legal metrology Department

l. Liquid effluent discharge

m. Tariff Advisory Committee Guidelines

n. Requirements of other authorities concerned with the Project

SECTION-IV
TENDER DRAWINGS & DOCUMENTS FOR REFERENCE

TENDER DRAWINGS & DOCUMENTS FOR REFERENCE

SL NO	DESCRIPTION	DOCUMENT NUMBER	REVISION
1	Plot Plan	90-0RG2N-PL-001	L
2	Process Plant Layout	20-0RG2N-BE-011	C
3	Process Plant Building	20-0RG2N-BE-001	A
4	Low Flash Additive Storage	20-0RG2N-BE-019	B

SECTION-V
TECHNICAL SPECIFICATIONS

TECHNICAL SPECIFICATIONS

SL NO	DESCRIPTION	DOCUMENT NUMBER
1	Functional Specification for AS/RS Warehouse (including ASRS Annexures)	ANNEX-1 (50-0RG2N-IS-006)
2	Functional Specifications for RFID based Trace-N-Track system	ANNEX-2
3	ASRS – Throughput calculations for sizing	ANNEX-3
4	Specification for Dock Levellers	ANNEX-4
5	Specification for Goods Lifts	ANNEX-5
6	Specifications for Telescopic conveyors / Rough Top Belt conveyor	ANNEX-6
7	Specification for Fork Lift Trucks (Battery Operated)	ANNEX-7
8	Specification for Fork Lift Trucks (Diesel Powered)	ANNEX-7
9	Specifications for Electric Pallet trucks	ANNEX-8